

5(a)(i)	1. $N\lambda$	B1
	2. N/f	B1
5(a)(ii)	v (= distance / time) = $N\lambda / (N/f)$ so $v = f\lambda$	B1
5(b)	$T = 4.0 \times 0.20 = 0.80$ (ms) or 8.0×10^{-4} (s)	C1
	$f = 1 / 8.0 \times 10^{-4}$ $= 1300$ Hz	A1
5(c)(i)	constant phase difference (between the waves)	B1
5(c)(ii)	180°	A1
5(c)(iii)	path difference = 2λ or $S_1Y - S_2Y = 2\lambda$	C1
	distance = $7.40 + (0.85 \times 2)$ $= 9.1$ m	A1